Mohammed Abu Baker (Shahoyi)

AI Research Scientist | LLM evaluation, interpretability and safety
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Profile

Research scientist in large language model (LLM) evaluation, interpretability and safety. Visiting Researcher and Principal Investigator on a £230K Coefficient Giving grant at Meridian, Cambridge; from Oct 2026, will lead a 4-person team on a $1M Coefficient Giving grant (planned base: London Initiative for Safe AI, LISA). Equal-contribution first author at COLM 2026. Four years of production software engineering. MSc AI (Distinction).

Research Experience

Visiting Researcher and Principal Investigator, Meridian, Cambridge, UK | Oct 2025 – Jul 2026

- Set a new state of the art on AuditBench: an investigator agent using my perplexity-differencing tool detected hidden behaviours in 74% of cases (previous best tool 59%; no tool 38%).
- Led a £230K Coefficient Giving grant (Oct 2025 – Jul 2026) showing most model organisms (0.5B–70B parameters) are "leaky"; COLM 2026 paper with two co-authors.

Research Assistant, Anglia Ruskin University, Cambridge, UK | May 2025 – Jul 2025

- With co-authors, collected and released a computer vision re-identification dataset of 1,232 images and 1,233 videos of 206 great crested newts, with boxes, masks and evaluation splits.
- Showed that Swin Transformer, convolutional (CNN) and foundation models rely on background and posture: on a debiased split, top-1 accuracy fell from 93.3% to 6.7% or lower.

Research Projects

Predicting fine-tuning side-effects before training (ML Alignment and Theory Scholars (MATS) 10, Feb 2026 – Mar 2026) | slides: docs.google.com/presentation/d/1QBw1AKbvHidJ2vwtVnFKUK5mQmkKuodeJ8YRqNM7M9w

- With a collaborator, steered Qwen2.5 with "intention vectors" from dataset activations; this induced emergent misalignment before any training, detectable from a single data point.

MSc thesis: fuzzing LLMs to elicit hidden behaviours (2026) | arxiv.org/abs/2606.29646

- Injected Gaussian noise into the weights and activations of 6 backdoored LLMs (7B–13B); fuzzing beat temperature sampling on 4 of 6 models, by up to about 6x.
- Selected fuzzing hyperparameters with Thompson sampling on a cheap proxy task, raising activation-fuzzing elicitation about 4x over a uniform sweep.

Sleeper-agent attention analysis (2025) | github.com/mshahoyi/sa_attn_analysis

- Fine-tuned backdoored Qwen2.5-3B models in Python (PyTorch, Hugging Face Transformers); ablation, activation patching and KL divergence localised attention changes to layers 20–30.

Encoded-reasoning model organism (Alignment Research Engineer Accelerator (ARENA) 6.0, Sep 2025 – Oct 2025) | lesswrong.com/posts/uznhavfcivA8vojMS

- With teammates, trained a model with reinforcement learning (RL) to induce encoded reasoning; it did not emerge, and we published the negative result.

Publications

* equal contribution

- Abu Baker, M.*, Baroni, L.*, Wilhelm, D.* Model Organisms Are Leaky: Perplexity Differencing Often Reveals Finetuning Objectives. Conference on Language Modeling (COLM), 2026. openreview.net/forum?id=LG1jPxGG7t
- Abu Baker, M., Babu-Saheer, L. Fuzzing Large Language Models to Elicit Hidden Behaviours. COLM 2026 Actionable Interpretability Workshop. arxiv.org/abs/2606.29646

- Abu Baker, M., Babu-Saheer, L. Mechanistic Exploration of Backdoored Large Language Model Attention Patterns. arXiv preprint, 2025. arxiv.org/abs/2508.15847
- Rosa, M. J., Abu Baker, M., Tang, C. M., Littlemore, J. GCN-ID: A Benchmark Dataset for Great Crested Newt Re-Identification Using AI Foundation Models. BMVC 2025 Machine Vision for Climate Change Workshop.
  bmva-archive.org.uk/bmvc/2025/assets/workshops/MVCC/Paper_13/paper.pdf; dataset: zenodo.org/records/18000809

Industry Experience

CEO and Founding Engineer, Kurdivia (later Padash), Erbil, Iraq | Aug 2021 – Sep 2024

- As founding engineer, built the real-time backend of a live trivia app in Node.js (NestJS) and scaled it to 150K concurrent players and 700K monthly active users on AWS EKS, Kafka and Redis.
- Led a team of 13 (3 engineers, 1 designer, 9 operations staff) and grew the app to 1.4M downloads; the company still operates, at $0.55M annual recurring revenue (Aug 2026).

Full-Stack Developer, NassWallet, Erbil, Iraq | Dec 2020 – Jul 2021

- Cut First Contentful Paint by 65% (from about 20s to 7s), and cut dashboard development time by 70% with a self-built module generator.

Education

MSc Artificial Intelligence, Anglia Ruskin University, Cambridge, UK | Jan 2025 – May 2026
Distinction (75.6%); top student across all master's programmes in the Computing department (MSc AI cohort: 45). Thesis: Fuzzing Large Language Models to Elicit Hidden Behaviours.

Bachelor of Medicine, Hawler Medical University, Erbil, Iraq | Sep 2014 – May 2020
Ranked 4th of 149 (75.5%).

Skills

Machine learning: Python, PyTorch, Hugging Face Transformers, Pandas; LLM fine-tuning, evaluation and auditing; steering vectors; reinforcement learning (RL); mechanistic interpretability (activation patching, ablation); computer vision
Software and infrastructure: TypeScript, Node.js, NestJS, PostgreSQL; AWS EKS, Kubernetes, Kafka, Redis, GitHub Actions, ArgoCD
Languages: Kurdish (native), English (fluent, IELTS 7.5), Turkish (fluent), Arabic (conversational), Persian (understanding)